Adaptive coupling between equation-based and agent-based models — A case study in epidemiological modeling —

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Abstract

This talk will demonstrate the potential breakthrough constituted by the possibility implemented in GAMA[2] by the authors to simultaneously use equation-based and agent-based models when simulating large populations of agents. A case study in epidemiology will depict how to dynamically shift from one model to the other with respect to inner indicators, and how to implement upscaling and downscaling processes in order to benefit from the best of the two modeling methods.

Keyword: Equation-based modeling, Agent-Based Modeling, Simulation, GAMA, Epidemiology

1. Equation-based, agent-based modeling

Equation-based models are often used to predict the long-term dynamics of the systems that are being modeled. However, the variability between individuals belonging to these systems is difficult to represent, making these models suitable only under the strong hypothesis that the populations are large and homogeneous. Agent-based models, on the other hand, are based on a fine grain representation of the attributes and behavior of individuals and therefore provide a greater level of details, but they fail in gaining analytical insight into the behavior of the whole population.

2. Dynamic coupling

These approaches have often been compared, but rarely used simultaneously in a dynamic fashion, firstly because this coupling raises a number of questions regarding how to define the upscaling (from agents to equations) and downscaling (from equations to agents) processes required, secondly because of a lack of suitable tools. Following the work of the authors in enhancing the GAML language proposed in the GAMA [2] platform to support modelers in seamlessly using equation-based definitions of agents' behaviors, we show how to write a coupled version of a classical epidemiological model.

3. "switching" SIR model

In this SIR (susceptible-infected-recovered) model, people are represented by agents when the density of infected people is low, and by equations when it reaches a certain threshold, thereby composing a dynamically switching model where the best of both approaches can be exploited. Upscaling and downscaling processes are discussed, for instance on how to best compensate for the loss of information on spatial structures when downscaling.

References

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